THE LAWS OF SOFTWARE TEAMS

The Ringelmann effect

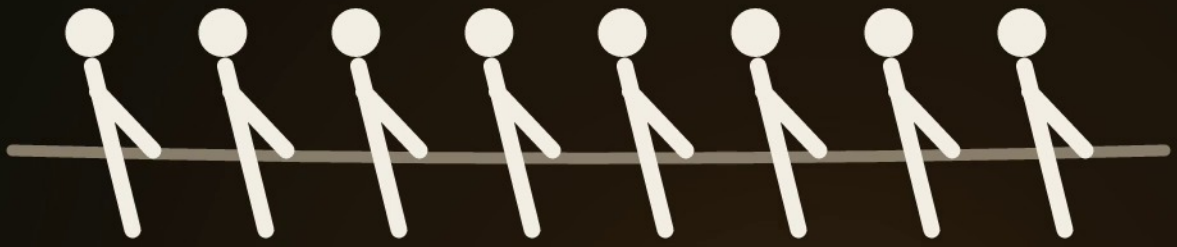
Individual effort drops as a group gets bigger

swipe →

Ivan Rykov

01 THE HOOK

8 people pulled a rope
They pulled like 4



total pull, in people

of 8

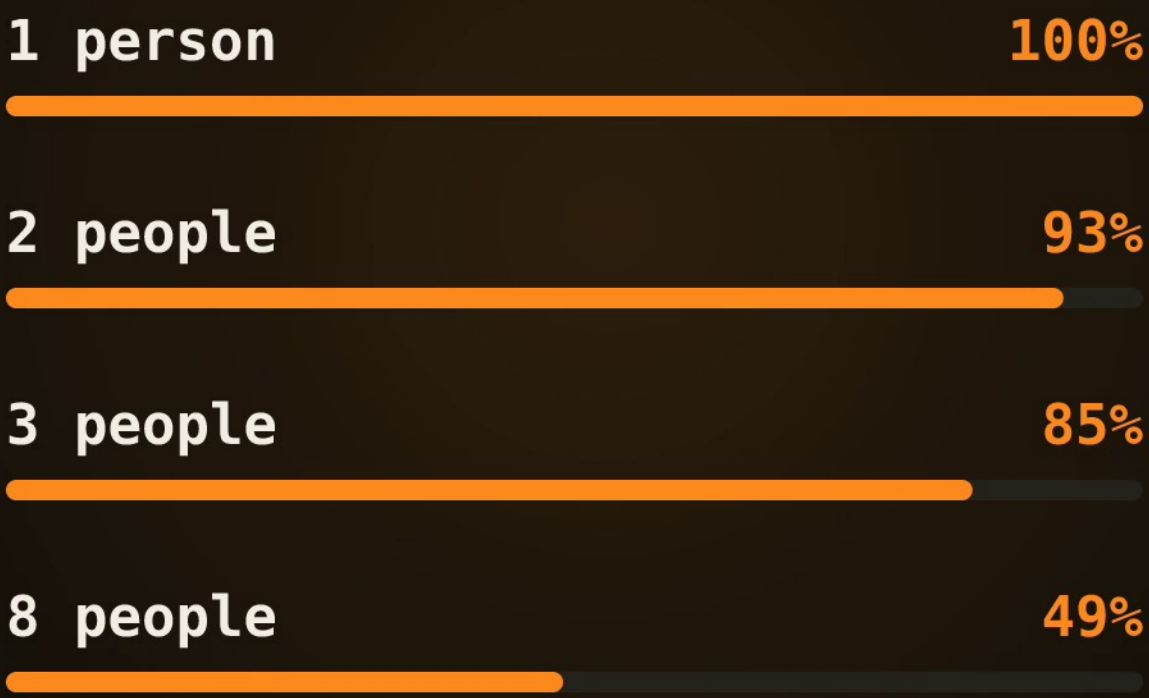
3.9

8 × 49% ≈ 3.9 · Ringelmann, 1913

Max Ringelmann

French agricultural engineer
Tested in the 1880s, published 1913

effort per person, by group size



Two things go wrong

1

Coordination

Nobody pulls at the same moment



2

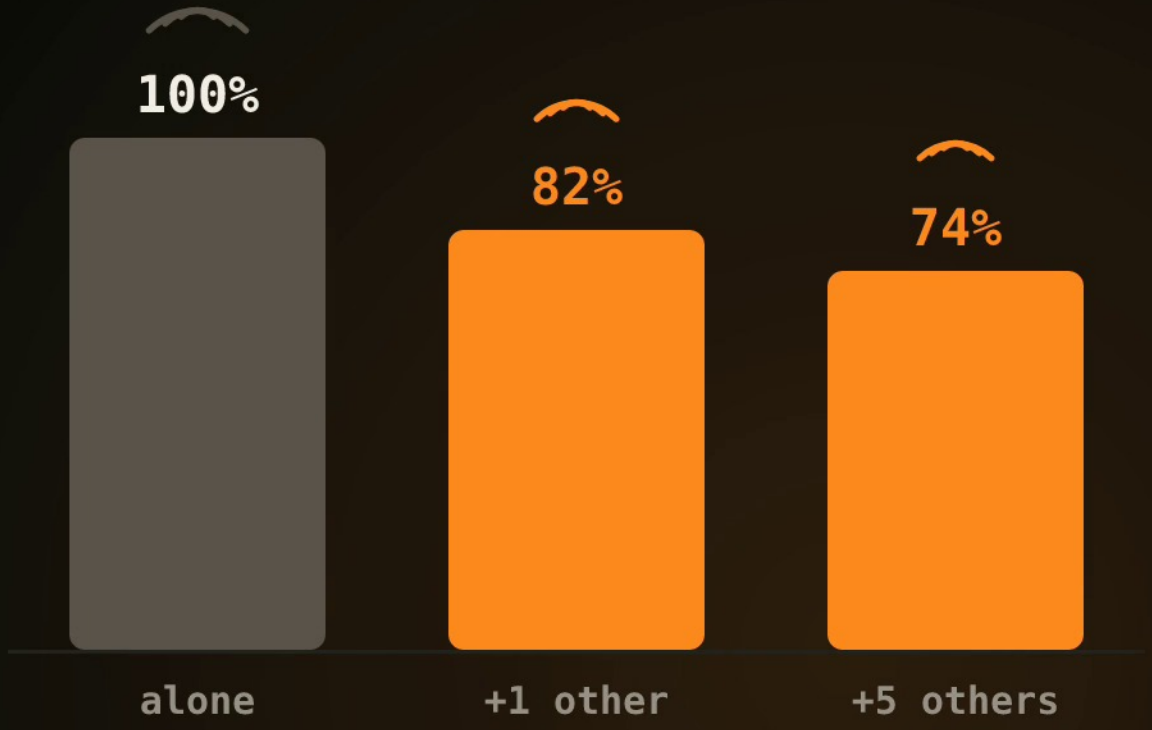
Motivation

“Someone else will pull”



1979: people shouted alone, blindfolded

But told others were shouting too:

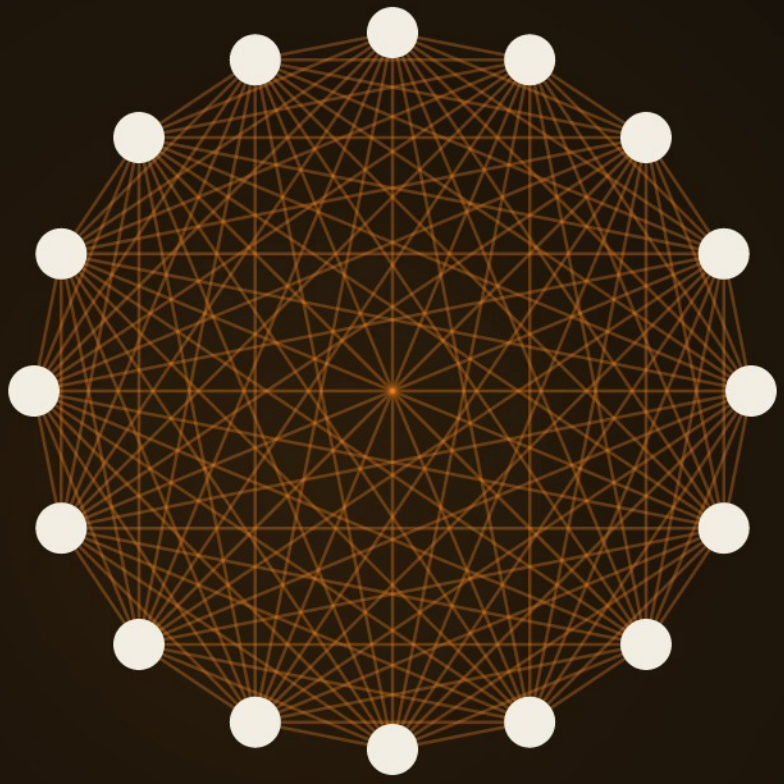


Nobody to coordinate with
They just tried less

Latané, Williams & Harkins, 1979 · “social loafing”

05 IN SOFTWARE

In software, everyone talks to everyone



16 people → 120 lines

n(n−1)/2 · Brooks, The Mythical Man-Month, 1975

The data agrees

390 software projects, same size:
teams of 9+ vs teams under 4

−30% schedule

+350% cost

+500% defects

Doug Putnam, QSM, 2018

How to beat it:

1. **Keep teams small**
   3-7 people, clear ownership

2. **Make each person's work visible**
   loafing hides in the crowd

3. **Give work that matters**
   effort follows meaning

Karau & Williams, 1993 · review of 78 studies

PostHog: 47 people, 15 teams
via manager.dev

So what's the right size? Small enough that nobody can hide



WHAT I TAKE FROM THIS

So I keep teams small and each person's work visible:

1. **The team that works best has three to five engineers who build the product, ideally four**
2. **It is time to split a team when coordination grows a lot, people slow down and the work gets harder to divide. Another sign: people work on unrelated parts of the product and are not interested in listening to each other**
3. **Everyone has their own area of responsibility. Even inside one feature, there is an informal feature lead who oversees the technical side**

WHAT I TAKE FROM THIS

4. A shared task with no owner and no clear priority gets lost. So every shared task goes to one named person who owns a part of the product, a feature or a release
5. People share what they did and how, and 1:1s are mandatory. This is also how warning signs show early: vague "work in progress", taking less responsibility, unusually many promises that are not kept, or "I help everyone" with nothing tracked
6. When I see these signs, I first talk to the person to understand why. They may be busy with other work or may have outgrown the role. They may have lost interest, or they may want to be useful to everyone and miss their own deadlines

WHAT I TAKE FROM THIS

7. **If they hide behind the team and let others do the work, I make their own area clearer, with no crowd around it. In a small team, hiding shows at once. The team itself will not tolerate it. If someone can hide in a team, the team is probably too big**

One more sign that it is time to split a team: the daily stand-up takes more than 15 minutes, and people do not remember who has already spoken

Sources: Kravitz and Martin, 1986 · Latané, Williams and Harkins, 1979 · QSM, Doug Putnam, 1997 and 2018

next: Price's law